DSC5001: Statistical Machine Learning I

Week 7: Flexible Regression: Local Smoothing and Gaussian Processes

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Semester A, 2026/27

Assignment Submission Format

Handwritten assignments are welcome.

- Scan every page clearly and upload the scanned document.
- **Do not submit ordinary photographs** of handwritten pages.

If you choose to use Word:

- Use Word's **Equation Editor** to typeset mathematical expressions.
- Avoid improvised formulas made from underscores, superscripts, or plain text.
- If equations are difficult to format, **handwrite and scan instead**.

Academic Integrity: AI Use

Assignment answers must be your own work.

- **Do not use AI to write assignment answers or explanations.**
- Show your own reasoning and calculations.
- During spot checks, I noticed a few submissions that raised concerns about possible AI-generated work. I have already reminded the students concerned.

Main idea

Please submit work that you can explain independently.

Group Project

The project deadline is about six weeks away.

- Continue developing your project over the remaining weeks.
- **In-class presentation:** November 30, 2026
- The presentation takes place in the **final teaching week**.
- **Every group must present** in class.

Use the remaining time to finalize the question, model comparison, validation, results, and presentation.

Final Examination

The planned examination structure is

40 multiple-choice questions

20 calculation-focused
apply methods and formulas

20 concept-focused
explain and interpret ideas

There will also be **two written-response questions**: choose **one** to answer.
You may bring up to **two cheat sheets**; both sides of each sheet may be used.

Exam Preparation

Before the final examination, I will provide **sample questions** to show the expected question style and difficulty.
The main revision materials are:

- the lecture slides and material covered in class;
- the sample questions.

Learning Objectives

By the end of this lecture, you should be able to:

1. build and interpret local regression using kernels and bandwidth;
2. explain why local linear regression can improve local prediction;
3. explain Gaussian process regression and predictive uncertainty;
4. compare flexible regression methods using validation evidence.

Roadmap

Part 1 Local smoothing and Nadaraya–Watson regression
Part 2 Local linear regression and bandwidth selection
Part 3 Gaussian process regression
Part 4 Choosing among regression methods

Main idea

Both local regression and Gaussian processes use similarity between inputs, but they use it in different ways.

Local smoothing

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Regression Goal

Let X denote a predictor and Y a response.
We observe training data

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n},$$

where n is the number of observations.
Our target is the conditional mean

$$f(x) = \mathbb{E}[Y \mid X = x].$$

At a new input x_0, we want to estimate

$$f(x_0).$$

Why Use Local Information?

Suppose we predict electricity demand from temperature.
For a target day with

$$x_0 = 30^\circ\mathrm{C},$$

which historical observation should be more informative?

$$29^\circ\mathrm{C} \qquad \text{or} \qquad 10^\circ\mathrm{C}?$$

Usually the $29^\circ\mathrm{C}$ observation should matter more.

Main idea

Local regression gives more influence to observations near the target input.

Global and Local Regression

A global model fits one curve for the full input range, for example

$$\widehat{f}(x) = \widehat{\beta}_0 + \widehat{\beta}_1 x + \widehat{\beta}_2 x^2.$$

Splines use global basis functions to make this curve more flexible.
A local method instead predicts at each target x_0 using nearby observations more strongly.

different target x_0 $\Rightarrow$ different local weights

Bandwidth

We know nearby observations should matter more, but we still need to decide what "nearby" means.
The bandwidth

$$\boxed{h > 0}$$

sets the neighborhood scale around the target x_0.

	Small h	**Large** h
Neighborhood	narrow	wide
Fit	more local	smoother
Variance	usually higher	usually lower

Main idea

Bandwidth controls how quickly influence decreases as we move away from x_0.

Smoothing Kernel

Bandwidth sets the neighborhood width. We still need a rule that converts distance into influence.
A smoothing kernel is a nonnegative function

$$K(u) \geq 0,$$

where the scaled distance is

$$u_i = \frac{x_i - x_0}{h}.$$

The raw weight of observation i is

$$\boxed{w_i(x_0) = K\left(\frac{x_i - x_0}{h}\right).}$$

A larger kernel value means more influence on the prediction.

Triangular Kernel

A simple kernel used in our numerical examples is

$$\boxed{K(u) = \max(1 - |u|, 0).}$$

For example,

$$K(0) = 1, \qquad K(0.5) = 0.5, \qquad K(1) = 0.$$

Thus observations with $|u| \geq 1$ receive zero weight.
Other choices include Gaussian and uniform kernels; bandwidth usually has the larger practical effect.

Kernel Weights and Bandwidth

Use $K(u) = \max(1 - |u|, 0)$ with inputs

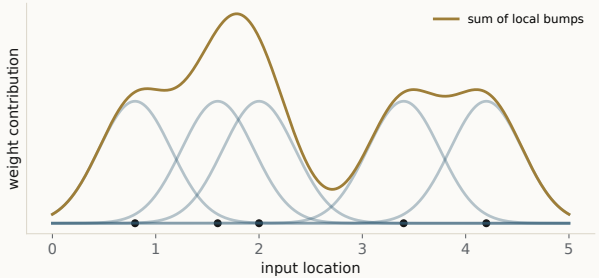
$$x = (1, 2, 3, 4, 5), \qquad x_0 = 3.$$

The raw weights are $w_i = K((x_i - x_0)/h)$.

Bandwidth	Scaled distances	Raw weights
$h = 2$	$(-1, -0.5, 0, 0.5, 1)$	$(0, 0.5, 1, 0.5, 0)$
$h = 1$	$(-2, -1, 0, 1, 2)$	$(0, 0, 1, 0, 0)$

A smaller h makes the estimate depend on fewer nearby inputs.

Kernel Bumps



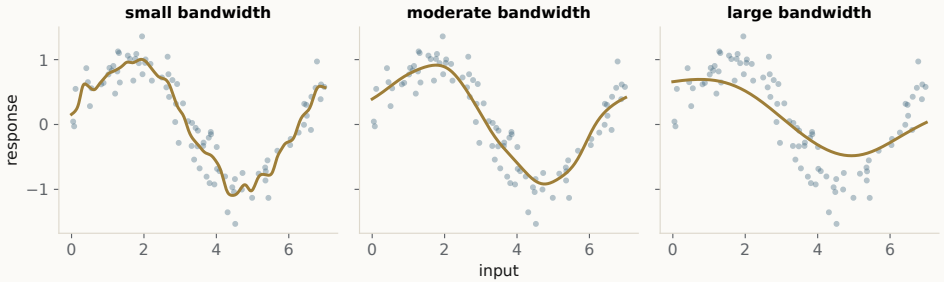
Black marks
input locations

Blue bumps
individual local weights

Gold curve
combined weight

Takeaway. Each observation has its strongest local influence near its own input location.

Bandwidth and Smoothness



Small h
more local

Moderate h
intermediate

Large h
smoother

Takeaway. Bandwidth controls how strongly the fitted curve reacts to individual observations.

From Weights to a Prediction

When the total raw weight is positive, normalize the weights to form an average:

$$\alpha_i(x_0) = \frac{w_i(x_0)}{\sum_j w_j(x_0)}, \qquad \sum_i \alpha_i(x_0) = 1.$$

For responses $y = (3, 6, 9)$ and weights $w = (1, 2, 1)$,

$$\alpha = (0.25, 0.50, 0.25).$$

The local prediction is

$$\boxed{\widehat{f}(x_0) = 0.25(3) + 0.50(6) + 0.25(9) = 6.}$$

Kernel Regression Example

Goal: predict the response at $x_0 = 1$ from

$$x = (0, 1, 3), \qquad y = (2, 5, 8),$$

using

$$h = 2, \qquad K(u) = \max(1 - |u|, 0).$$

The scaled distances are

$$\frac{x_i - x_0}{h} = \left(-\frac{1}{2}, 0, 1\right),$$

so

$$w = (0.5, 1, 0).$$

Since $\sum_i w_i = 1.5$,

$$\boxed{\alpha = \left(\frac{1}{3}, \frac{2}{3}, 0\right).}$$

Kernel Example: Prediction

Continue the same example:

$$y = (2, 5, 8), \qquad \alpha = \left(\frac{1}{3}, \frac{2}{3}, 0\right).$$

Using

$$\widehat{f}_h(x_0) = \sum_i \alpha_i(x_0) y_i,$$

we obtain

$$\widehat{f}_h(1) = \frac{1}{3}(2) + \frac{2}{3}(5) + 0(8) = \boxed{4}.$$

The point at $x = 3$ does not affect this prediction because its kernel weight is zero.

From Averaging to Local Fitting

Previously, $y = (2, 5, 8)$ and $w = (0.5, 1, 0)$ gave the prediction 4.
Why use this weighted average? Near $x_0 = 1$, try a single prediction a and measure its weighted squared error:

$$Q(a) = 0.5(2 - a)^2 + (5 - a)^2 + 0(8 - a)^2.$$

The best local constant is the value minimizing this error:

$$\boxed{\widehat{a} = \arg\min_a Q(a).}$$

The next calculation shows why that value is the weighted average.

Local Constant Solution

For target x_0, minimize

$$Q(a; x_0) = \sum_i w_i(x_0)(y_i - a)^2.$$

Differentiate and set the derivative to zero:

$$-2 \sum_i w_i(x_0)(y_i - a) = 0 \implies \sum_i w_i(x_0) y_i = a \sum_i w_i(x_0).$$

If the weights are nonnegative and $\sum_i w_i(x_0) > 0$, the unique solution is

$$\boxed{\widehat{a} = \frac{\sum_i w_i(x_0) y_i}{\sum_i w_i(x_0)}.}$$

Nadaraya–Watson Estimator

The best local constant is therefore a weighted average.
Choose the weights from the kernel:

$$w_i(x_0) = K\left(\frac{x_i - x_0}{h}\right).$$

Substituting gives

$$\boxed{\widehat{f}_h(x_0) = \frac{\sum_i K\left(\frac{x_i - x_0}{h}\right) y_i}{\sum_i K\left(\frac{x_i - x_0}{h}\right)}.}$$

Nadaraya–Watson regression is local constant weighted least squares.

Effective Sample Size

A prediction can be dominated by only a few observations, even with a large dataset.
For normalized weights $\alpha_i(x_0)$, a measure of weight concentration is

$$\boxed{n_{\text{eff}}(x_0) = \frac{1}{\sum_i \alpha_i(x_0)^2}.}$$

For $\alpha = (0.2, 0.6, 0.2)$,

$$\boxed{n_{\text{eff}} = 1/(0.04 + 0.36 + 0.04) \approx 2.27.}$$

A compact-support kernel may also give *zero total weight* in an empty neighborhood; then increase h or avoid unsupported prediction.

Boundary Motivation

A local constant works best when the neighborhood is reasonably balanced around x_0.
Near a boundary, most nearby observations may lie on only one side, so a local average can be pulled toward the interior.

Interior target	**Boundary target**
points on both sides	points mainly on one side
local average often balanced	local average may be biased

Example. At the left boundary $x_0 = 0$, use

$$x = (0, 1, 2), \qquad y = (1, 3, 5), \qquad w = (1, 0.5, 0.1).$$

These points follow $f(x) = 1 + 2x$, so

$$\boxed{f(0) = 1}.$$

Boundary Example: Local Constant

Same example:

$$x = (0, 1, 2), \qquad y = (1, 3, 5), \qquad x_0 = 0, \qquad w = (1, 0.5, 0.1).$$

The local constant estimate is

$$\widehat{f}_{\mathrm{LC}}(0) = \frac{1(1) + 0.5(3) + 0.1(5)}{1 + 0.5 + 0.1} = \boxed{1.875}.$$

But the underlying local trend gives

$$f(0) = 1.$$

The one-sided observations pull the weighted average upward.

Local Linear Model

Instead of one constant near x_0, allow a local slope:

$$\boxed{y_i \approx a + b(x_i - x_0).}$$

Here a is the fitted local level and b is the local slope.
Because the centered input $x - x_0 = 0$ at the target, the prediction is

$$\boxed{\widehat{f}(x_0) = \widehat{a}.}$$

We fit a new weighted line when the target x_0 changes.

Local Linear Objective

Choose a and b to minimize

$$\boxed{\sum_i w_i(x_0)\,[y_i - a - b(x_i - x_0)]^2\,.}$$

This is weighted least squares applied locally.
A new local line is fitted for each target x_0.

Main idea

The local slope lets the model follow a nearby trend instead of only averaging response values.

Matrix Form

Let $u_i = x_i - x_0$ be the centered input, $w_i = w_i(x_0)$ its local weight, and $\boldsymbol{y} = (y_1, \ldots, y_n)^\top$. Define

$$\boldsymbol{X}_0 = \begin{bmatrix} 1 & u_1 \\ \vdots & \vdots \\ 1 & u_n \end{bmatrix}, \qquad \boldsymbol{W}_0 = \operatorname{diag}(w_1, \ldots, w_n).$$

Let

$$\boldsymbol{\gamma} = \begin{bmatrix} a \\ b \end{bmatrix}.$$

Then local linear regression is a weighted least-squares problem.

Local Linear Solution

The weighted least-squares solution is

$$\boxed{\widehat{\boldsymbol{\gamma}} = (\boldsymbol{X}_0^\top \boldsymbol{W}_0 \boldsymbol{X}_0)^{-1} \boldsymbol{X}_0^\top \boldsymbol{W}_0 \boldsymbol{y}}$$

when

$$\boldsymbol{X}_0^\top \boldsymbol{W}_0 \boldsymbol{X}_0$$

is invertible.
The prediction at x_0 is the first component:

$$\widehat{f}(x_0) = \widehat{a}.$$

Boundary Example: Local Line

Same goal: predict at $x_0 = 0$ using a local line.
Since

$$u_i = x_i - x_0 = (0, 1, 2),$$

the local design and weight matrices are

$$\boldsymbol{X}_0 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \boldsymbol{W}_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.1 \end{bmatrix}.$$

Boundary Example: Solve the Line

For the same $x = (0, 1, 2)$, $y = (1, 3, 5)$, and $w = (1, 0.5, 0.1)$, solve

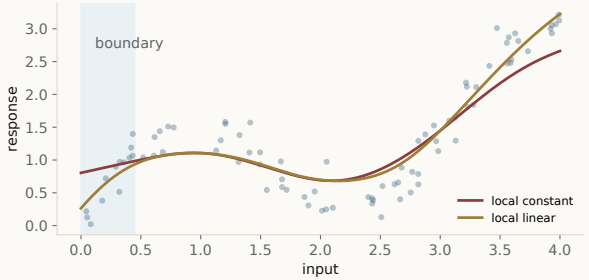
$$\begin{bmatrix} 1.6 & 0.7 \\ 0.7 & 0.9 \end{bmatrix} \begin{bmatrix} \widehat{a} \\ \widehat{b} \end{bmatrix} = \begin{bmatrix} 3 \\ 2.5 \end{bmatrix}.$$

This gives $\widehat{a} = 1$, $\widehat{b} = 2$, so

$$\boxed{\widehat{f}_{\mathrm{LL}}(0) = 1.}$$

The earlier local constant estimate was 1.875. The local line recovers the trend $f(x) = 1 + 2x$ in this example.

Boundary Behavior



Dots
observations

Red curve
local constant

Gold curve
local linear

Takeaway. Near a boundary, a local line can use the local trend instead of only averaging one-sided responses.

Local Polynomial Regression

A local line is often enough, but sometimes the relationship bends within a neighborhood. Then fit a local polynomial of degree q:

$$y_i \approx a_0 + a_1(x_i - x_0) + \cdots + a_q(x_i - x_0)^q.$$

Important cases are

$$q = 0 \quad \text{local constant}, \qquad q = 1 \quad \text{local linear}.$$

Bandwidth h remains the main control of smoothness.

Choosing the Bandwidth

A very small h may follow noise.
A very large h may smooth away real structure.
Therefore bandwidth should be selected using predictive evidence rather than visual preference.

choose h using validation or cross-validation

Leave-One-Out Cross-Validation

For each observation i:

1. remove observation i;
2. fit the smoother using the other $n-1$ observations;
3. predict at x_i;
4. compare the prediction with y_i.

Let

$$\widehat{f}_{h,-i}(x_i)$$

denote this leave-one-out prediction.
Then

$$\boxed{\mathrm{CV}(h) = \frac{1}{n}\sum_i \left[y_i - \widehat{f}_{h,-i}(x_i)\right]^2.}$$

Bandwidth Selection Example

Suppose

h	0.1	0.3	0.8	1.5
CV MSE	2.9	1.7	1.2	1.8

The smallest CV error is

$$1.2.$$

Therefore,

$$\boxed{h = 0.8}$$

is selected from these candidates.
After selection, refit using the available training data before final evaluation.

Locality in Higher Dimensions

With p features, an observation is $\boldsymbol{x}_i \in \mathbb{R}^p$, and the target is $\boldsymbol{x}_0$. We can measure closeness using $\|\boldsymbol{x}_i - \boldsymbol{x}_0\|_2$.
Suppose a box-shaped neighborhood covers half the range in each feature:

Number of features p	1	2	5
Volume fraction	0.5	$0.5^2 = 0.25$	$0.5^5 = 0.03125$

In high dimensions, a local region can contain very little data. This is the **curse of dimensionality**.

Feature Scaling

Suppose two features are age and annual income.
Compare

$$(30,\ 50000)$$

with

$$(40,\ 51000).$$

The raw differences are

$$10 \quad \text{and} \quad 1000.$$

Without scaling, income dominates Euclidean distance simply because its numerical units are larger.
A common solution is to standardize numerical features using training-set statistics.

Gaussian process regression

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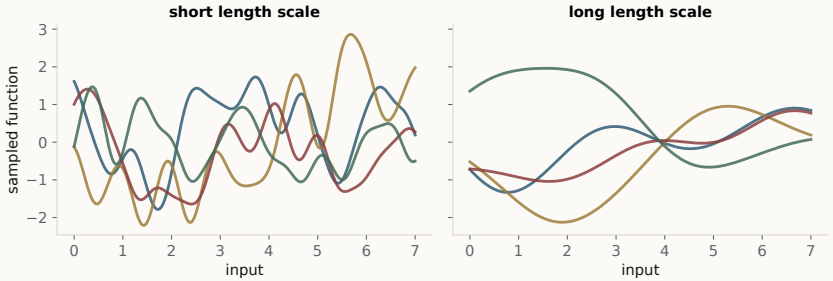
Why Gaussian Processes?

Suppose we predict electricity demand from outdoor temperature.
A local smoother gives a predicted value. But we may also ask:

How reliable is that prediction?

For example, predictions near observed temperatures are often more reliable than those far from the data.
A **Gaussian process (GP)** estimates both a regression curve and uncertainty about that curve.

Prior Function Samples



Each curve
one possible relationship

Short length scale
changes rapidly

Long length scale
changes smoothly

Takeaway. A GP starts with many plausible curves, not one fixed fitted curve.

Prior and Posterior

A GP describes what functions are plausible *before* and *after* observing data.

Prior: before data	**Posterior: after data**
many possible curves	curves updated by observed responses
initial uncertainty	revised uncertainty

The observations change both our best prediction and how uncertain we are about it.

What Is a GP?

A GP is a probability model for an unknown function:

$$\boxed{f \sim \mathcal{GP}(m, k).}$$

- $f(x)$: unknown mean response at input x;
- $m(x)$: expected value of $f(x)$ before observing data;
- $k(x, x')$: how the two function values vary together.

At any finite collection of inputs, the possible function values follow a **joint Gaussian distribution**.

Covariance Means Connection

The GP covariance function is

$$\boxed{k(x, x') = \mathrm{Cov}(f(x), f(x')).}$$

For two similar temperatures, a large positive covariance means their hidden demand levels tend to move together.

Large covariance	**Small covariance**
stronger connection	weaker connection

Covariance is **not** the probability that two function values are equal.

RBF Covariance

We need a rule to assign covariance to any pair of inputs.
A common choice is the **RBF covariance**:

$$\boxed{k(x, x') = \sigma_f^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right).}$$

- $|x - x'|$: distance between the two inputs;
- σ_f^2: prior variance of the hidden function;
- $\ell > 0$: length scale controlling how fast covariance falls.

This k models *covariance*; the earlier K assigned local averaging weights.

RBF: A Numerical Example

Use the RBF covariance with $\sigma_f^2 = 1$ and $\ell = 1.2$.

$$k(x, x') = \exp\left(-\frac{(x - x')^2}{2(1.2)^2}\right).$$

Input distance $\lvert x - x' \rvert$	0	1	2
Covariance	1	0.707	0.249

The numbers measure how strongly the hidden function values are related, *not* probabilities.

Length Scale

Keep the input distance at one unit and $\sigma_f^2 = 1$.

Length scale ℓ	0.6	1.2	2.4
$k(x, x+1)$	0.249	0.707	0.917

- Small ℓ: covariance falls quickly; more rapid changes are plausible.
- Large ℓ: covariance falls slowly; smoother curves are favored.

Function and Measurement

Our observations contain more than the underlying regression function:

$$\boxed{y_i = f(x_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_n^2).}$$

- $f(x_i)$: hidden mean at input x_i;
- y_i: actual measured response;
- ε_i: independent Gaussian observation noise;
- σ_n^2: noise variance.

Here $\mathcal{N}(0, \sigma_n^2)$ means a normal distribution with mean zero and variance σ_n^2.

Why a Zero Prior Mean?

For our example, let the response Y be **electricity demand above a fixed baseline**, measured in demand units.

- $Y = +2$: demand is two units above the baseline.
- $Y = -1$: demand is one unit below the baseline.

Before observing data, we can use $m(x) = 0$: no expected deviation from the baseline.
If the baseline is 100 units and the GP predicts a deviation of $+1.13$, the predicted total demand is 101.13 units.

One Observation: Setup

Goal: predict the mean demand deviation at $30°\mathrm{C}$.
We observe one day at $29°\mathrm{C}$ with a deviation of $+2$ units:

$$x_1 = 29, \quad x_* = 30, \quad y_1 = 2.$$

Take an RBF GP with $m(x) = 0$, $\sigma_f^2 = 1$, $\ell = 1.2$, and noise variance $\sigma_n^2 = 0.25$.

$$\underbrace{k(29, 30)}_{\text{prior covariance}} \approx 0.707, \qquad k(29, 29) = k(30, 30) = 1.$$

The new input x_* plays the same role as the target x_0 in local regression.

One Observation: Prediction

Same example: $y_1 = 2$, covariance to target 0.707, and noise variance 0.25.
A GP combines the prior and this observation:

$$\underbrace{\mu_*}_{\text{predicted mean}} = \frac{k(x_1, x_*)}{k(x_1, x_1) + \sigma_n^2}\, y_1.$$

Substitute the numbers:

$$\boxed{\mu_* \approx \frac{0.707}{1 + 0.25}(2) = 1.13.}$$

We predict a demand deviation of about $+1.13$ units at $30^\circ\mathrm{C}$.
It is below $+2$ because the new input is different and the observation contains noise.

One Observation: Uncertainty

Same target: $x_* = 30$, with prior variance 1.
The GP also calculates how much uncertainty remains:

$$\underbrace{s_*^2}_{\text{variance of hidden } f(x_*)} = k(x_*, x_*) - \frac{k(x_1, x_*)^2}{k(x_1, x_1) + \sigma_n^2}.$$

Numerically,

$$\boxed{s_*^2 \approx 1 - \frac{0.707^2}{1.25} = 0.60.}$$

	Before data	**After data**
Mean	0	1.13
Function variance	1	0.60

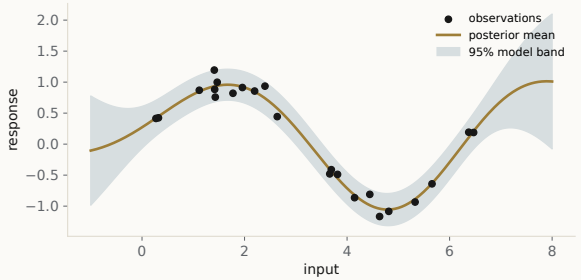
Near and Far from Data

Keep the measurement $y_1 = 2$ at $29^\circ\mathrm{C}$.

Target	Predicted deviation	Function variance
$30^\circ\mathrm{C}$ (near)	$+1.13$	0.60
$36^\circ\mathrm{C}$ (far)	approximately 0	approximately 1

Far away, the RBF covariance with the observed input is almost zero.
The GP then returns toward its prior mean and prior uncertainty.

Observed Data Update the GP



Black dots
observations

Gold curve
updated mean

Blue band
function uncertainty

Takeaway. The mean curve changes after observing data, and uncertainty often shrinks near observations.

Reading the GP Band

The blue band describes uncertainty about the *hidden mean function*.

- Near informative data, the band is often narrower.
- Far from data, it often widens toward the prior level.
- Its width depends on the GP assumptions and chosen parameters.

A narrow band is not proof that the fitted model is correct.

Two Types of Uncertainty

The hidden function value and a future measurement are not the same thing:

$$y_* = f(x_*) + \varepsilon_*, \qquad \varepsilon_* \sim \mathcal{N}(0, \sigma_n^2).$$

If the GP gives function variance s_*^2, then

$$\boxed{\mathrm{Var}(y_* \mid \text{data}) = s_*^2 + \sigma_n^2.}$$

A future measurement contains fresh noise, so its prediction interval is wider.

Prediction Intervals: Numbers

Return to the one-observation example at $30^\circ\mathrm{C}$.
We found $\mu_* \approx 1.13$, $s_*^2 \approx 0.60$, and $\sigma_n^2 = 0.25$.

Quantity	Variance	Approx. 95% interval
Hidden mean $f(x_*)$	0.60	$[-0.39, 2.65]$
Next response y_*	0.85	$[-0.68, 2.94]$

We use $\mu_* \pm 1.96\sqrt{\text{variance}}$. Both intervals refer to demand *deviation* from the baseline.

Two Observations: Setup

Goal: predict the mean demand deviation at $29^\circ\mathrm{C}$, now using *two* measured days.

Temperature	$28^\circ\mathrm{C}$	$30^\circ\mathrm{C}$
Observed deviation	$+1$	$+2$

Use the same RBF settings:

$$\sigma_f^2 = 1, \quad \ell = 1.2, \quad \sigma_n^2 = 0.25, \quad m(x) = 0.$$

This time $n = 2$, $\boldsymbol{y} = (1, 2)^\top$, and $x_* = 29$.

Two Observations: Covariances

Same example: target $29^\circ\mathrm{C}$; observations at $28^\circ\mathrm{C}$ and $30^\circ\mathrm{C}$.
From the RBF formula:

$$k(28, 30) \approx 0.249, \qquad k(28, 29) = k(30, 29) \approx 0.707.$$

The noisy training covariance and target covariance vector are

$$\boxed{\boldsymbol{C} = \begin{bmatrix} 1.25 & 0.249 \\ 0.249 & 1.25 \end{bmatrix}}, \qquad \boxed{\boldsymbol{k}_* = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}}.$$

The diagonal $1.25 = 1 + 0.25$ includes observation noise; the target's prior variance is $k_{**} = 1$.

GP Notation for Several Data Points

We have n observed responses $\boldsymbol{y} = (y_1, \ldots, y_n)^\top$ and one target x_*.

$\boldsymbol{K}$	prior covariances among training function values
$\boldsymbol{C} = \boldsymbol{K} + \sigma_n^2 \boldsymbol{I}$	covariance of noisy training responses
$\boldsymbol{k}_*$	covariances between training inputs and target
k_{**}	prior function variance at the target

Here $\boldsymbol{I}$ is the $n \times n$ identity matrix. The numbers on the previous slide are an example with $n = 2$.

GP Formula for Many Observations

Assume a zero-mean GP and independent Gaussian observation noise.
The posterior distribution of $f_* = f(x_*)$ is

$$\boxed{f_* \mid \boldsymbol{y} \sim \mathcal{N}(\mu_*, s_*^2),}$$

where

$$\boxed{\mu_* = \boldsymbol{k}_*^\top \boldsymbol{C}^{-1} \boldsymbol{y}} \qquad \text{(predicted mean)},$$

$$\boxed{s_*^2 = k_{**} - \boldsymbol{k}_*^\top \boldsymbol{C}^{-1} \boldsymbol{k}_*} \qquad \text{(remaining function variance)}.$$

These come from conditioning a joint Gaussian distribution; no long derivation is needed here.

Two Observations: Calculation

Continue the target-29 example:

$$\boldsymbol{C} = \begin{bmatrix} 1.25 & 0.249 \\ 0.249 & 1.25 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{k}_* = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}.$$

To avoid calculating $\boldsymbol{C}^{-1}$ explicitly, solve

$$\boldsymbol{C}\boldsymbol{a} = \boldsymbol{y}, \qquad \boldsymbol{C}\boldsymbol{b} = \boldsymbol{k}_*.$$

The solutions are approximately

$$\boxed{\boldsymbol{a} = \begin{bmatrix} 0.501 \\ 1.500 \end{bmatrix}}, \qquad \boxed{\boldsymbol{b} = \begin{bmatrix} 0.471 \\ 0.471 \end{bmatrix}}.$$

Two Observations: Prediction

Continue the target-29 example: $k_{**} = 1$, $\boldsymbol{k}_* \approx (0.707, 0.707)^\top$.
From the previous slide, $\boldsymbol{a} \approx (0.501, 1.500)^\top$ and $\boldsymbol{b} \approx (0.471, 0.471)^\top$.

$$\mu_* = \boldsymbol{k}_*^\top \boldsymbol{a} \approx 0.707(0.501 + 1.500) = \boxed{1.41},$$
$$s_*^2 = k_{**} - \boldsymbol{k}_*^\top \boldsymbol{b} \approx 1 - 0.707(0.471 + 0.471) = \boxed{0.334}.$$

The predicted demand is 1.41 units above the baseline, with remaining uncertainty about the hidden mean.

GP Hyperparameters

The GP prediction depends on three important settings:

- ℓ length scale: how quickly the curve may change
- σ_f^2 function variance: how much the hidden curve may vary
- σ_n^2 noise variance: how noisy measurements are

These are called **hyperparameters**. Different settings can produce different curves and uncertainty bands.

Choosing GP Settings

One approach is to compare prediction errors on held-out data.
For example, keeping other settings fixed, suppose validation gives:

Length scale ℓ	0.6	1.2	2.4
Illustrative validation MSE	1.2	0.7	1.1

Here $\boxed{\ell = 1.2}$ is best among the candidates.
Another approach, **marginal likelihood**, selects parameters by how plausible they make the observed responses under the GP model.

GP Computational Cost

GP calculations use a covariance matrix with n rows and n columns for n training observations. For exact GP regression, typical costs are

$$\boxed{\text{memory } O(n^2), \qquad \text{factorization } O(n^3).}$$

Roughly, doubling n makes the dense matrix four times larger and the main factorization about eight times more expensive.
Large datasets often require approximate GP methods.

When Can a GP Be Wrong?

A narrow posterior band means the *fitted GP* is confident under its assumptions.
It is not a guarantee that the real relationship lies inside that band.
Check whether:

- the covariance function describes the actual trend;
- the assumed noise level is realistic;
- the training inputs cover the prediction region.

Use validation and model diagnostics alongside GP uncertainty.

When Are GPs Useful?

GP regression can be attractive when both prediction and uncertainty matter.
Examples:

- predicting environmental or scientific measurements;
- planning expensive experiments;
- deciding which input to measure next (Bayesian optimization).

A GP gives a curve and a model-based estimate of how uncertain that curve is.

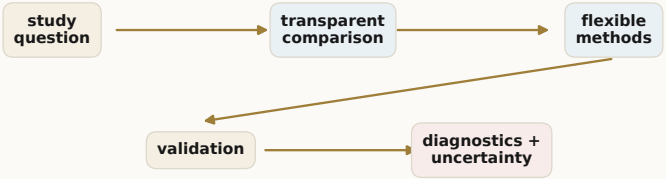
Choosing a regression method

DSC5001 ▪ Semester A, 2026/27

Methods at a Glance

Method	Main control	Useful when
Linear regression	features	a global linear trend is adequate
Polynomial / spline	degree, knots	smooth nonlinear structure is expected
Local regression	bandwidth	nearby observations carry useful local information
Gaussian process	covariance parameters	prediction and uncertainty are both important

One Regression Workflow



Use the same data split and evidence standards for every candidate method.

Takeaway. Define the question, compare candidate models fairly, and then interpret the selected fit.

When Scores Are Similar

Suppose validation gives

Model	Validation MSE
Linear regression	2.40
Cubic spline	1.72
Local linear regression	1.68
Gaussian process	1.70

The three flexible models perform similarly.
A very small MSE difference is not the only consideration.

Interpretability
prefer simpler models when useful

Uncertainty
GP is useful when uncertainty matters

Computation
avoid unnecessary complexity

Summary

Local regression	uses nearby observations; bandwidth controls locality
Local linear regression	adds a local slope and can improve boundary behavior
Gaussian process	models a distribution over functions and provides predictive uncertainty
Model choice	use validation first, then consider interpretation, uncertainty, and cost

There is no universally best regression model.

Week 7: flexible regression from local weighting to probabilistic functions

Questions and discussion